Contact

www.linkedin.com/in/ddunn-tech (LinkedIn)

Top Skills

Vector Databases
Hugging Face Transformers
Fine-Tuning Pipelines

Drew Dunn

Senior AI Engineer & Computational Scientist | Applied AI/ML, Scientific Computing & Quantitative Modeling | Healthcare & Fintech
United States

Summary

Senior AI/ML Engineer with a strong foundation in distributed systems, data platform architecture, and production-grade software engineering. My career has evolved across enterprise software, financial systems, and healthcare technology, consistently focusing on systems that process, model, and operationalize data at scale.Expertise spans the full machine learning lifecycle and modern backend architectures, including Python (Pydantic/Mypy), TensorFlow, PyTorch, Apache Spark, Databricks, and MLOps automation. Having built my career around complex, regulated data environments, my recent work heavily focuses on advanced Generative AI architectures, Agentic workflows, and designing rigorous evaluation frameworks and adversarial testing strategies to stress-test frontier LLM logic.I bridge the gap between academic research and production engineering. Whether architecting high-performance RAG pipelines or scaling distributed data workflows, I prioritize data integrity, clean type-safe architecture, and building reliable, observable AI systems aligned with measurable business impact.

Experience

Elevance Health
Senior AI Engineer & Computational Scientist
November 2023 - Present (3 years)

I bridge high-level math and machine learning with production software engineering. I design and launch machine learning models and AI systems that solve complex healthcare problems—from digging through messy clinical and claims data to building predictive risk engines and working with Large Language Models. Having a background in physics and computer science allows me to approach these massive health datasets with mathematical rigor, ensuring our models aren't just theoretically sound, but fast, reliable, and actually useful in production.

Cognizant

Senior Machine Learning Engineer

October 2020 - October 2023 (3 years 1 month)

Focused on building end-to-end machine learning pipelines for enterprise clients. I designed feature engineering workflows, trained predictive models, and deployed scalable inference services. I also implemented model monitoring systems and automated retraining pipelines, ensuring production reliability. My work bridged data engineering and applied machine learning, enabling real-time and batch inference use cases.

Citi

Senior Data & Machine Learning Engineer

August 2017 - September 2020 (3 years 2 months)

Working in a highly regulated financial environment, I built resilient systems for transaction processing and risk analytics. My responsibilities extended beyond backend engineering into large-scale data engineering, where I designed data ingestion workflows and batch-processing systems for financial modeling. I collaborated with quantitative analysts to productionize statistical models used in risk assessment and fraud detection.

Epicor

6 years 2 months

Senior Data Engineer

August 2016 - July 2017 (1 year)

I led architectural improvements for distributed systems supporting global customers. I introduced structured data pipelines and improved system observability. My responsibilities included mentoring engineers and driving modernization efforts, including early experimentation with predictive modeling techniques embedded into enterprise workflows.

Data Engineer

September 2013 - August 2016 (3 years)

I played a key role in designing scalable application layers and integrating data-intensive modules across enterprise platforms. My work increasingly involved data transformation, ETL logic, and system integrations that required reliability and performance at scale. During this period, I began implementing statistical modeling techniques for internal forecasting features, blending software engineering with applied data science.

Full Stack Developer

June 2011 - September 2013 (2 years 4 months)

I expanded into full-stack development while contributing to backend services that processed high-volume business transactions. I collaborated with database architects to improve query optimization and data consistency. I also supported early internal analytics initiatives, building reporting pipelines that transformed operational data into structured insights.

Epicor
Software Engineering Co-op
June 2009 - January 2010 (8 months)

Contributed to internal development projects, assisting in feature implementation, testing, and documentation. Gained practical experience in enterprise software environments and exposure to real-world development workflows, including source control, code reviews, and collaborative engineering practices.

Education

The College of Saint Rose
Master of Science · (May 2015)

The College of Saint Rose
Bachelor of Science · (May 2011)

Texas Tech University
Doctor of Philosophy · (September 2022)